

Input-files of makeCutawayForGMT

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File name	Contents
<i>Arbitrary name</i>	Controlling parameters
mesh.dat	Mesh information
resistivity_block_iterX.dat *	Information of parameter cells

* In the file name, ‘X ‘ indicates the iteration number.

How to use makeCutawayForGMT

You need to execute the following command in the directory where input files exist.

```
makeCutawayForGMT [Name of parameter file] [-aniso]
```

By default, the files for the isotropic inversion are assumed.

To use the output files of anisotropic inversion, the argument '-aniso' is needed after the name of the parameter file.

File format of parameter file

Mesh type (0: Tetrahedral mesh, 1: Brick mesh , 2: Deformed non-conforming hexahedral element)

Iteration number

Type of cross section (0: Vertical cross-section, 1: Horizontal cross-section)

X (km)

Y (km)

Z (km)

Rotation center

Rotation angle around the z-axis (deg.) ¹⁾

1) Please see the next slide

Number of the parameter cells excluded from cross section (N_{EX})

Index of the 1st parameter cell excluded from cross section

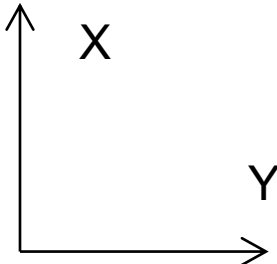
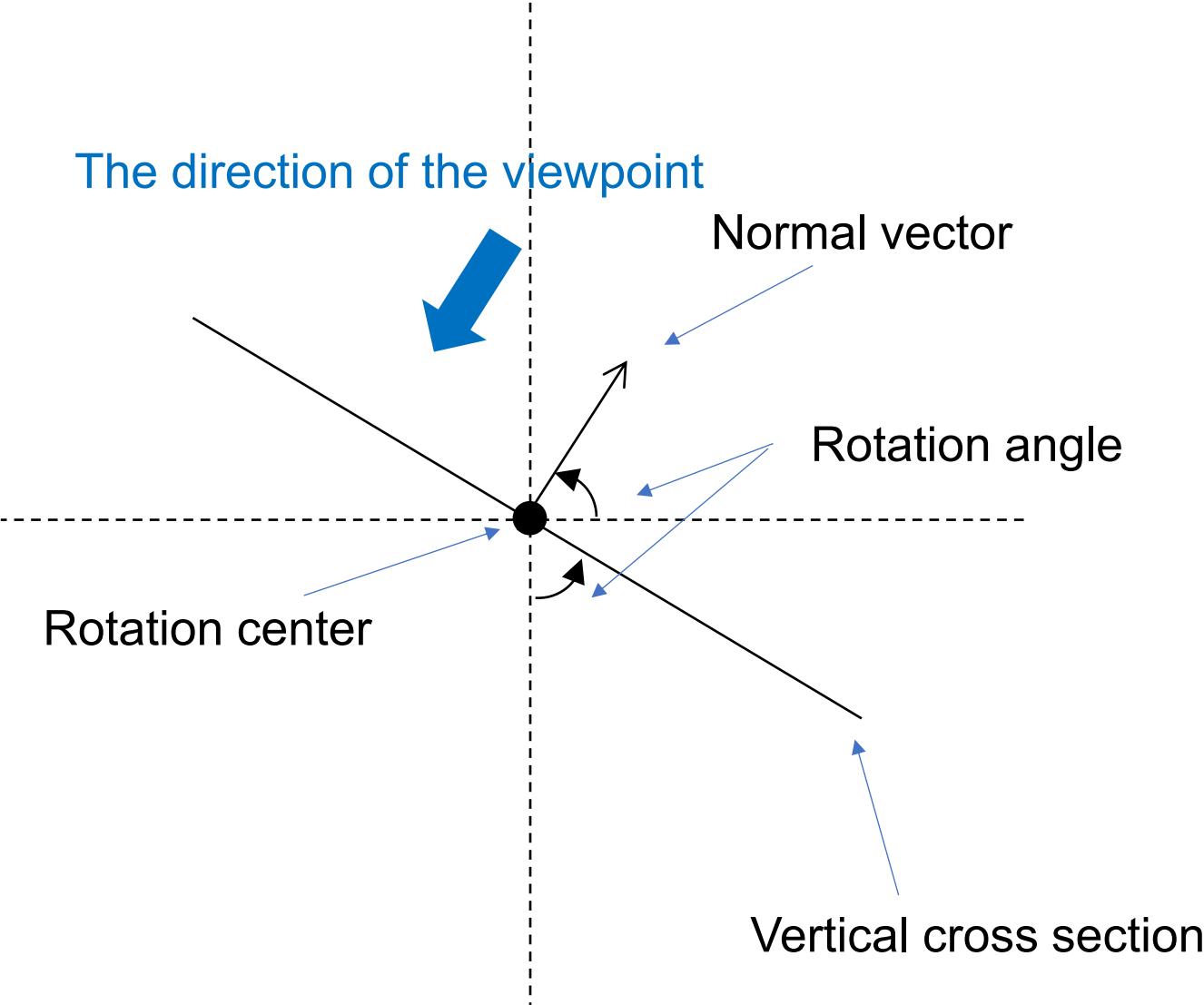
⋮

Index of the N_{EX} -th parameter cell excluded from cross section

Threshold for anisotropic rotation angle ²⁾

2) The threshold is used only for the anisotropic case (when the argument ‘-aniso’ is used). If the $|\log(\rho_{xx}/\rho_{yy})|$ is lower than the threshold, rotation angles (θ_S , θ_D , and θ_L) of the corresponding area are not outputted.

Rotation angle



When the rotation angle is zero, the cross section is made on the ZX-plane.

Output-files of makeCutawayForGMT (1/2)

Isotropic case (by default)

File name	Contents
resistivity_GMT_iterX.dat	Log resistivity values on the cross section. It can be used as an input file of 'psxy' command of GMT.

In the file name, 'X ' indicates the iteration number.

<Example>

```
> -Z1.354416e+00
-2.500000e+03 0.000000e+00
-1.500000e+03 0.000000e+00
-1.500000e+03 5.000000e-03
-2.500000e+03 5.000000e-03
-2.500000e+03 0.000000e+00
```

Common logarithm of resistivity

Composing points of the polygon that has the above resistivity.

```
> -Z1.354416e+00
-1.500000e+03 0.000000e+00
-9.987360e+02 0.000000e+00
-9.987360e+02 5.000000e-03
-1.500000e+03 5.000000e-03
-1.500000e+03 0.000000e+00
```

By default, there is no space between '-Z' and the resistivity value, following the format of GMT version 5 or higher.

When using GMT version 4 or lower, please set 'isGMT5OrHigher = false' in 'CommonParameters.h' to add spaces.

Output-files of makeCutawayForGMT (2/2)

Anisotropic case (when the argument '-aniso' is used)

File name	Contents
rhoXX_GMT_iterX.dat	$\log(\rho_{xx})$ on the cross-section.
rhoYY_GMT_iterX.dat	$\log(\rho_{yy})$ on the cross-section.
rhoZZ_GMT_iterX.dat	$\log(\rho_{zz})$ on the cross-section.
strike_GMT_iterX.dat	θ_S on the cross-section.
dip_GMT_iterX.dat	θ_D on the cross-section.
slant_GMT_iterX.dat	θ_L on the cross-section.
anisotropy_GMT_iterX.dat	$ \log(\rho_{xx}/\rho_{yy}) $ on the cross-section.

In the file name, 'X ' indicates the iteration number.

It can be used as an input file of 'psxy' command of GMT.

By default, there is no space between '-Z' and the resistivity parameters, following the format of GMT version 5 or higher. When using GMT version 4 or lower, please set 'isGMT5OrHigher = false' in 'CommonParameters.h' to add spaces.